



INSTRUCTIONAL LABORATORIES AND DEMONSTRATIONS

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A versatile Hong–Ou–Mandel interference experiment in optical fiber for the undergraduate laboratory

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Hong–Ou–Mandel (HOM) interference is a quantum optics laboratory experiment that has recently become more accessible to undergraduate students. The experiment consists of two identical photons simultaneously entering a nonpolarizing beamsplitter, where the photon wavefunctions interfere. As a result, the photon pair exits from the same beamsplitter output, whereas classically, the two photons are equally likely to exit from the two different outputs. This effect is demonstrated by observation of a dip in coincidence counts measured between the two beamsplitter outputs. Due to the precision needed to achieve photon indistinguishability, the setup and alignment of this experiment is often considered to be too difficult and time consuming to be appropriate for the undergraduate instructional laboratory. Here, we present an alternative optical-fiber-based apparatus that gives a consistently reproducible experiment. In our approach, quantum interference occurs within a fused-fiber coupler, instead of within a traditional beamsplitter. We use a commercially available fiber-coupled biphoton source that requires minimal alignment and increases the interference coherence length. In addition, our biphoton source provides direct temperature-based control of the frequency degeneracy of the photon pairs, allowing students to investigate physical properties of HOM interference such as coherence length and interference visibility. Through use of standard optomechanical parts, combined with the commercially available fiber-integrated biphoton source, our apparatus is positioned midway between a completely built-from-scratch and a pre-aligned setup, making it ideal for the advanced instructional laboratory. © 2025 Published under an exclusive license by American Association of Physics Teachers.

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I. INTRODUCTION

The nature of interference in quantum mechanics is a source of confusion for many physics students. One phenomenon that demonstrates quantum interference is the Hong–Ou–Mandel (HOM) effect, which was first reported experimentally by Hong *et al.* in 1987.¹ In the HOM effect, two individual photons enter different input ports of a 50:50 nonpolarizing beamsplitter and, if the two photons have identical frequency, polarization, arrival time, and position in space, they are quantum mechanically indistinguishable and HOM interference occurs. When this situation is viewed classically, one expects that the two photons will exit from different beamsplitter output ports 50% of the time. However, due to quantum interference between identical photons, the photon pair is more likely to exit from the same output port. Thus, if the experiment is set up to detect coincidence counts between the two output ports, few to no coincidences will be measured because the two photons “bunch” at

a single output. The observation of this dip in coincidence counts makes HOM interference a very intuitive demonstration of a quantum optics effect.

HOM interference is realizable in an undergraduate laboratory;^{2–4} however, it is usually implemented as an instructor-assembled demonstration because alignment of an HOM apparatus can be difficult and time consuming for an undergraduate student to perform. In order to achieve indistinguishable photons, the two photons input to the beamsplitter must be aligned horizontally and vertically so that they overlap when exiting the beamsplitter. In addition, the two photons must enter the beamsplitter almost simultaneously so that their spatial modes overlap. Furthermore, the common method of producing photon pairs using spontaneous parametric down conversion in β -barium borate crystals produces photon pairs that have an HOM coherence length on the order of $10\text{ }\mu\text{m}$,^{1,3} which is a level of precision that often requires tedious alignment steps at the beamsplitter before aligning the actual interference beams. Due to these

many requirements, investigating anything more than initial interference is typically too intensive of a project for most undergraduate teaching labs to find worthwhile.

We address these challenges by presenting an apparatus that not only allows for easier alignment, but also provides opportunities to investigate other physical properties of HOM interference. Through the use of both commercially available pre-aligned parts as well as parts that must be aligned by the student, we find a middle ground between buying a fully off-the-shelf setup and having to meticulously align every part of a built-from-scratch apparatus.

Advantages of our approach are as follows. First, the majority of the apparatus consists of fiber optic cables, which minimizes the alignment procedures necessary. Second, we employ a fiber-integrated photon-pair source that produces pairs with coherence lengths on the order of hundreds of microns, making effective simultaneity easier to achieve as well as eliminating the tedious alignment of the traditional down-conversion setup. Third, the photon-pair source can be tuned to adjust the difference in frequency between the two photons produced as well as the rates of photon-pair production. This control allows more experimental parameters to be investigated in relation to the behavior of HOM interference. On the other hand, the remainder of the apparatus must be built and aligned by the student giving them valuable experience with an optomechanical setup. The combination of commercially available and do-it-yourself parts makes our apparatus ideal for a longer advanced undergraduate experiment as students can perform the full set up, alignment, and measurements over the course of five to eight weeks.

II. APPARATUS

The interfering photon pairs are produced using a phenomenon known as spontaneous parametric down conversion (SPDC). SPDC occurs when a high-power pump laser illuminates a crystal with nonlinear optical properties. Pump photons are converted into two photons known as the *signal* and *idler*, with the requirement that $\omega_i + \omega_s = \omega_p$, where $\omega_{i,s,p}$ are the frequencies of the idler, signal, and pump photons, respectively. Because the signal and idler are produced at the same time, if two photons are detected in our setup within a sufficiently short window of time, we call this a coincidence and assume that the two photons we detected were those same signal and idler photons.

In our experiment, we produced photon pairs via collinear type II SPDC using a periodically poled potassium titanyl phosphate (PPKTP) nonlinear crystal.⁵ In collinear type II SPDC, the signal and idler are polarized orthogonally and emitted in the same direction of travel as the pump photon.

A diagram of the full apparatus is given in Fig. 1. The pump laser (Ondax LMFC-405) is a 405 nm laser diode coupled to single-mode polarization-maintaining (PM) fiber. All of the fiber used in the interference apparatus is single-mode and polarization-maintaining in order to prevent modal and polarization dispersion.⁶ The biphoton source (Qubitekk QES2.2) is coupled to single-mode PM fiber and is directly connected to the pump laser via a ferrule connector/physical contact (FC/PC) fiber mating sleeve (Thorlabs ADAFC2). This biphoton source consists of a temperature-controlled PPKTP crystal with a spectral filter to remove the pump beam from the output. The photon pairs produced have wavelengths close to 810 nm, with wavelengths of exactly 810 nm for indistinguishable output. The serial emulator

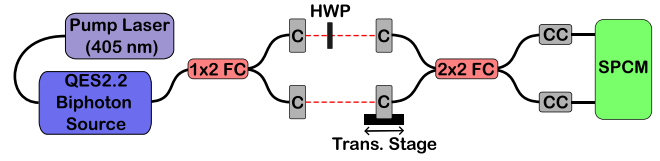


Fig. 1. A diagram of the interference apparatus. 1×2 FC is the one-input two-output fused-fiber coupler, C labels the 810 nm collimators, HWP is the 810 nm half-wave plate, 2×2 FC is the two-input two-output fused-fiber coupler, CC labels the coupler cages that filter visible light, and SPCM stands for single-photon counting module that records both single-photon and coincidence rates for the two fiber coupler outputs. The solid connecting lines in the diagram are single-mode polarization-maintaining fiber optic cables, and the dotted lines are paths of photons in air.

Termite⁷ is used to interface with the biphoton source and control the PPKTP crystal temperature. Due to the temperature dependence of PPKTP's optical properties,⁵ the temperature control in the biphoton source allows direct control of the frequency degeneracy of the output. The photon pairs are orthogonally polarized when they leave the biphoton source, upon which they enter a one-input two-output fused-fiber coupler (Thorlabs PFC780F)—labeled 1×2 FC in Fig. 1—that separates them between two outputs based on their polarization.

The two outputs of the 1×2 coupler are connected using FC/PC connectors to two collimators consisting of a fixed focus collimating lens (Thorlabs F280FC-780) mounted (Thorlabs KM100T) with XY tilt control. These two collimators start the in-air section of the apparatus. This section consists of two delay lines, one is fixed and one is adjustable. Details of the two delay lines are shown in Fig. 2. Each delay line has another identical collimator at its end. The fibers are polarization-maintaining fibers, and so the collimators are all oriented so that the photons leave and enter with vertical polarization and an 810 nm half-wave plate is inserted into the fixed delay line to precisely match the polarization in the adjustable delay line. Note that although the photons are initially orthogonally polarized, they both leave the 1×2 fused-fiber coupler vertically polarized due to the orientation of the connector keys at the outputs as the connector key is aligned with the slow axis of the fiber. The adjustable delay line is terminated with a collimator mounted to a translation stage (Thorlabs PT1), which adjusts the optical path length difference of the photon pair as they enter the 2×2 fiber cable. The translation stage has a resolution of 0.02 mm and a range of 25 mm. This resolution is precise enough to show the shape of the HOM dip, while the range is more than ample to capture the entire dip, which is on the order of 1 mm wide.

Because the output of the biphoton source is not visible light, it is easiest to first roughly align the two delay lines using a visible laser and maximizing counts through the delay lines. We use a 633 nm HeNe laser (Melles Griot 05-LHP-151) that is coupled into a multimode fiber (Thorlabs M31L01) using a collimator with both XY tilt and translation control consisting of a mount (Thorlabs K6XS) and fixed focus collimating lens (Thorlabs F220FC-B). More precise alignment is then done using the 810 nm photons.

The collimators at the end of the delay line are connected to a two-input two-output fused-fiber coupler (OF Link FPMC-810-22-50-PM-C-05-FP-B), which is labeled 2×2 FC in Fig. 1. Unlike the first coupler, this one takes the two vertically polarized photons and sends each with equal

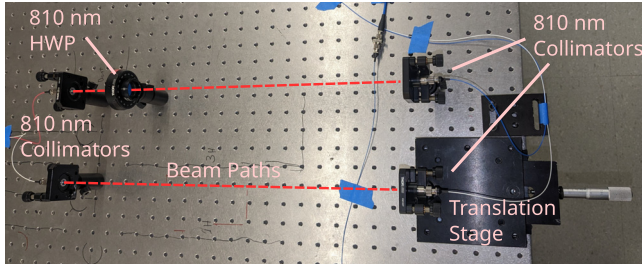


Fig. 2. A labeled picture of the fixed and adjustable delay lines with the beam paths as dashed lines. The 810 nm half-wave plate is in the fixed delay line. The translation stage is used to change the length of the adjustable delay line.

probability to either of the two outputs. This operation is the analog of the role of a nonpolarizing beamsplitter in other HOM setups. The outputs of the 2×2 coupler are connected to two coupler cages (labeled CC in Fig. 1) that use longpass filters (Thorlabs, FGL780) to filter out visible light before the remaining photons are sent to the single-photon counting module, abbreviated as SPCM (Excelitas SPCM-AQRH-13-FC), which creates a pulse that is sent to the field programmable gate array, or FPGA, (Digilent Nexys3 100 MHz system clock). The FPGA records both the counts at each output as well as the coincidences between them.

The data collected by the SPCM and FPGA counting module are sent to a computer running a LabVIEW program, which shows live plots of single-photon and coincidence rates. The program records the raw data to a comma separated values (CSV) file, while more detailed analysis is done in Excel or MATLAB.

III. THEORY

A. Fused-fiber coupling

HOM interference was first demonstrated in a nonpolarizing beamsplitter and the fiber-optics-based undergraduate laboratory demonstrations often use a beamsplitter to achieve the actual interference as well.^{1,2,8} However, in our setup, the HOM interference occurs within a fused-fiber coupler. It is therefore necessary to understand what exactly a fused-fiber coupler does to a photon at the quantum mechanical level in order to motivate the theory of the HOM effect.

Fused-fiber couplers are traditionally understood as an application of the evanescent electric field phenomenon.⁹ However, it is equivalent to consider the wavefunction of a photon in a single-mode fiber. A single-mode fiber consists of a core several microns across surrounded by a cladding of lower refractive index.⁶ This effectively forms a cylindrical potential well. Due to continuity requirements, the wavefunction extends slightly out of the core of the fiber into the cladding. When the cores of two fibers are brought very close together with only a small amount of cladding between them, the system turns into two potential wells with a thin barrier, where the wavefunction of a photon in one well can tunnel into the other. If the wavefunction remains in this potential, the probability density will oscillate back and forth between the two wells.¹⁰

A fused-fiber coupler then consists of two fibers, usually single mode, which are fused together in the middle to create the potential described earlier. By adjusting the length of the coupling region, a manufacturer can control the time that the

photon remains in this potential and effectively control the probability of finding the photon at either of the outputs. The fiber coupler is symmetric, so the probability of staying or switching fibers is the same for both inputs. From a quantum mechanical perspective, the action of a fused-fiber coupler on an input state is the same as the action of a beamsplitter on an input state.⁸ Therefore, just as with a beamsplitter, we can represent the action of the fused-fiber coupler with the unitary 2×2 matrix operator,

$$\hat{M} = \begin{pmatrix} t & r \\ -r & t \end{pmatrix}, \quad (1)$$

where t is the transmission coefficient corresponding to the probability amplitude of the photon remaining in the same input fiber after the beamsplitter, and r is the reflection coefficient or the probability amplitude of the photon being reflected into the second fiber after the beamsplitter. Together, $t^2 + r^2 = 1$. The negative sign on the bottom left r is added to represent a phase change that is required due to energy conservation.¹¹

B. HOM interference

Although our experiment occurs in a fused-fiber coupler, it is instructive to consider exactly what HOM interference is in a beamsplitter before considering a more mathematically rigorous treatment of the coupler interference. When two photons come to the two inputs (called as Input 1 and Input 2) of a 50:50 nonpolarizing beamsplitter, they can behave in one of the following four options:

- (1) The photon in Input 1 transmits, and the photon in Input 2 reflects, leading to both photons exiting Output 1.
- (2) The photon in Input 1 reflects, and the photon in Input 2 transmits, leading to both photons exiting Output 2.
- (3) Both photons transmit, and one exits Output 1 and the other exits Output 2.
- (4) Both photons reflect, and one exits Output 1 and the other Output 2.

Classically, we expect to see all four options equally often. However, when the photons are indistinguishable, HOM interference causes options 1 and 2 to become much more likely than options 3 and 4, which in turn leads to a decrease in coincidences measured between the two outputs.

In our experiment, we use a fused-fiber coupler which, from Eq. (1), behaves similarly to a beamsplitter. With the action of the coupler defined, the purely quantum mechanical phenomenon of HOM interference can be described. Similar to the formulation of Weihs *et al.*,⁸ the state of the two-photon system entering the coupler (Fig. 3) is described by the double integral

$$|\Psi_{in}\rangle = \iint d\omega' d\omega'' \zeta(\omega', \omega'') |1, \omega'\rangle |2, \omega''\rangle, \quad (2)$$

where

$$\zeta(\omega', \omega'') \propto \delta(\omega' + \omega'' - \omega_p) f_{\omega'_c}(\omega') f_{\omega''_c}(\omega''). \quad (3)$$

Here, $|1, \omega'\rangle$ represents a photon of frequency ω' entering the coupler at input 1, $f_{\omega'_c}(\omega')$ is the single-photon frequency distribution, and δ is the Dirac delta distribution, added to adhere to the constraint that $\omega' + \omega''$ is equal to the pump

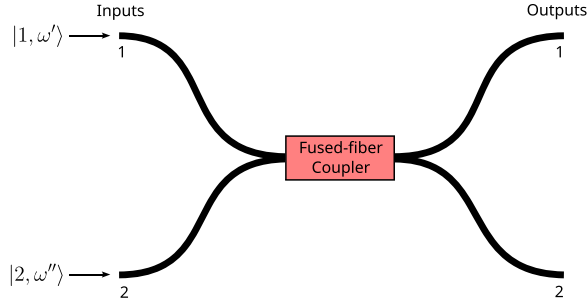


Fig. 3. The two-photon state enters the fused-fiber coupler. Here, a photon of frequency ω' enters at input 1 and a photon of frequency ω'' enters at input 2. The fused-fiber coupler then sends photon 1 to outputs 1 or 2 with probabilities t^2 or r^2 , respectively, and sends photon 2 to outputs 1 or 2 with probabilities r^2 or t^2 , respectively.

frequency ω_p . Although the SPDC in the PPKTP crystal produces a $\sin^2$ spectrum,¹² a Gaussian distribution can be used as an accurate approximation of the $\sin^2$ function.^{13–15} Therefore, $f_{\omega'_c}(\omega')$ is assumed to be a sharply peaked Gaussian distribution with a center frequency of ω'_c .

The interference is then given by the action of the matrix operator $\hat{M}$ on the photons at the two inputs,

$$|\Psi_{out}\rangle = (\hat{M} \otimes \hat{M})\Psi_{in}. \quad (4)$$

There is a more in-depth explanation of the action of such a matrix on the two-photon state in the Appendix as well as the calculation of outcome probabilities. Using the method mentioned in the Appendix and then integrating over all wavelengths, we can find probabilities of the different outcomes for the final state $|\Psi_{out}\rangle$. The details of these calculations can be found in Campos *et al.*¹⁶ and produce the following probabilities:

$$P_{out}(2, 0) = P_{out}(0, 2) = t^2 r^2 [1 + \Phi], \quad (5)$$

$$P_{out}(1, 1) = t^4 + r^4 - 2t^2 r^2 \Phi, \quad (6)$$

where

$$\Phi = \iint \zeta(\omega', \omega'') \zeta^*(\omega'', \omega') d\omega' d\omega''. \quad (7)$$

When the single-photon distributions are Gaussian as in our case, Φ is evaluated as¹⁶

$$\Phi = e^{-\omega_d^2 t_c^2 / 8} \times e^{-2\Delta x^2 / c^2 t_c^2}, \quad (8)$$

where c is the speed of light, $\omega_d = |\omega'_c - \omega''_c|$, t_c is the reciprocal of the spectral width of the two-photon distribution, and Δx is the optical path length difference between the photons arriving at the coupler. This equation assumes the photons to be identical in polarization. And so, Φ becomes a characterization of indistinguishability. When both $\omega_d = 0$ and $\Delta x = 0$, then $\Phi = 1$, which means the photons are quantum mechanically indistinguishable and $P_{out}(1, 1) = (t^2 - r^2)^2$ achieves full interference when $t^2 = r^2 = 1/2$. Similarly, as ω_d or Δx become much larger, Φ approaches 0, and the incoming photons begin to exit through different outputs as they are no longer indistinguishable and are able to behave in all four of the possible options given earlier. Plotting

$P_{out}(1, 1)$ as a function of Δx gives the characteristic “HOM dip,” an example of which is shown in Fig. 4.

Note that Eq. (8) has a factor dependent on ω_d as well as a factor dependent on Δx . Therefore, we can also plot a HOM dip with the relationship between coincidence counts and ω_d . We produce our photon pairs through collinear type II SPDC.¹⁷ This specific type of down conversion is highly temperature dependent with slight changes in the temperature of the down converting crystal affecting ω_d for the photon pairs produced.¹⁸ A unique aspect of this experimental setup is the ability to control the crystal temperature in the biphoton source.

We define the central temperature T_c to be the crystal temperature at which the photon pairs have the same wavelength or $\omega_d = 0$. Fluctuations around T_c will induce fluctuations in the width of the photon distribution ω_d . Defining $\Delta T = |T_c - T|$, we can say that $\Delta T \propto \omega_d$.¹⁸ Therefore, it is equivalent to plot a HOM dip with temperature as the independent variable. We carry out this procedure by measuring and plotting interference visibility at varying temperatures.

C. Treatment of accidentals

The data collection in this experiment relies heavily on counting coincidences between two detectors. The inherent flaw in this approach is that one cannot truly tell experimentally when two events are exactly simultaneous. Instead, we define a coincidence between two detectors to be when both detectors measure a count within the same time span, called the *coincidence window* and denoted τ_c . When the gap between individual counts at each detector is much larger than τ_c , this definition of coincidence is very effective at detecting actual instances of simultaneity. However, there will still be some *accidental coincidences* that do not come from photon pairs, but instead, due to randomness, result from unrelated photons arriving at each detector within the coincidence window and getting counted as a coincidence. The rate of accidental coincidences R_{acc} is given by¹⁹

$$R_{acc} = \tau_c R_A R_B, \quad (9)$$

where R_A and R_B are the count rates measured at detectors A and B, respectively. Using this relationship, we calculate accidental coincidence rates and subtract them from the overall data.

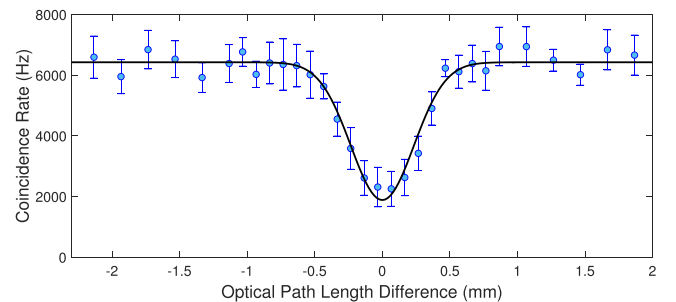


Fig. 4. Coincidence rate plotted against optical path length difference (OPD). Crystal temperature for the measurements is $(35.93 \pm 0.05)^\circ\text{C}$. Estimated accidental coincidences are subtracted from the raw coincidence counts and the error bars correspond to standard deviation of the averaged coincidence rates. The data are fitted to an inverted Gaussian distribution with equation $y = (6400 \pm 200)\text{Hz} - (4500 \pm 500)\text{Hz} \times e^{(-9 \pm 2)\text{mm}^{-2}x^2}$. This plot shows the characteristic HOM dip, where strong interference occurs when the OPD is equal to 0.

IV. SAMPLE DATA AND ANALYSIS

In collecting the sample data, coincidence counts and single counts were recorded using a LabVIEW program and analyzed using MATLAB. The LabVIEW program records the coincidence counts over a span of 0.3 s using a coincidence window of 10 ns and outputs the coincidence rate over that 0.3 s period. A 10 ns coincidence window was chosen to minimize the measured accidental coincidence rate R_{acc} , while still allowing enough time for photons to move through the beamsplitter. In our experiment, with R_A and R_B at approximately 150 000 counts per measurement period and a 10 ns coincidence window, we calculate a respectable accidental coincidence rate of around 200 counts per measurement period [from Eq. (9)]. For each data point in Fig. 4 at a particular optical path length, 200 individual coincidence rates were measured and then averaged after the subtraction of the accidental coincidence rate. For the dips in Fig. 5, each data point is the average of 100 coincidence rate measurements. The error bars represent the standard deviation of the mean for the distribution of measured values at each setting.

Merely demonstrating the existence of the dip is often the end goal of an undergraduate HOM experiment. An example of that result is shown in Fig. 4. Coincidence rates are measured in 0.1 mm increments near the point of zero optical path length difference and 0.2 mm increments outside the center range. After the accidental coincidence rates are subtracted out, the data are fitted to an inverted Gaussian distribution.

The depth of the dip, also known as the *interference visibility*, does not reach zero as would be expected for perfect indistinguishability. The main reason for this non-zero value is not that the photons are not indistinguishable; instead it is due to the fact that the two-input two-output fused-fiber coupler (where the interference occurs) does not have equal probability for sending a photon to either output. It can be shown using Eq. (6) that the visibility of the dip is given by

$$V = \frac{2r^2t^2}{r^4 + t^4}, \quad (10)$$

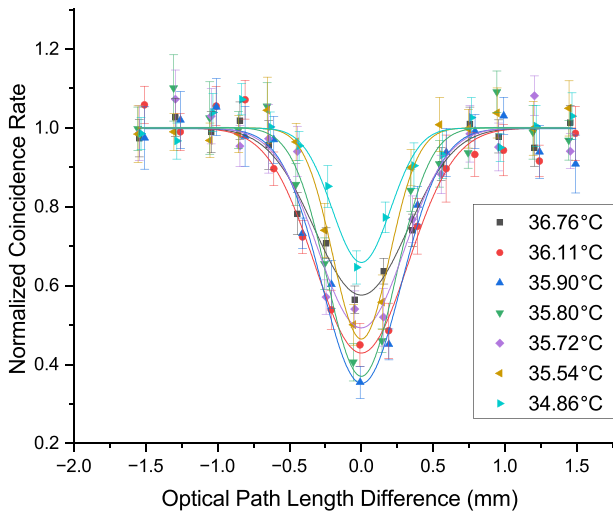


Fig. 5. Coincidence counts plotted against optical path length difference (OPD) for seven different crystal temperatures. The datasets are each fitted to an inverted Gaussian distribution as in Fig. 4 and are normalized with respect to the average coincidence counts outside of the dip with accidental coincidences removed.

where t^2 and r^2 are the transmission and reflection probabilities, respectively. Thus, $V = 0$ when there is no dip and $V = 1$ when the dip goes fully to zero coincidences. We measured the transmission and reflection of the fused-fiber coupler for indistinguishable photons and found that $t^2 = 0.466 \pm 0.003$ and $r^2 = 0.534 \pm 0.003$.

Here, Eq. (10) is a measurement of maximum dip visibility. It should be noted that while differences in transmission and reflection are one aspect affecting dip visibility, other imperfections in the fiber optic causing overlap in spatial modes and non-idealities in photon polarization likely affect the dip visibility as well.

Using Eq. (10), the maximum visibility that could be recorded using this apparatus is 0.991 ± 0.004 . The data of Fig. 4 does not quite reach this mark, but we do achieve interference visibility of 0.707 ± 0.072 , which passes the 50% mark necessary to distinguish classical and quantum HOM interference.²⁰

In addition to observing the HOM dip and confirming quantum interference, in Fig. 4, we confirm that the data follow an inverted Gaussian model as predicted. These findings are interesting on their own, but we can go further in investigating other characteristics of HOM interference.

By repeating the measurement process of Fig. 4 for multiple scenarios, we can investigate how frequency degeneracy affects the HOM dip. Because of the temperature control in the QES2.2 biphoton source, we are able to take measurements at nondegenerate wavelengths. In Fig. 5, coincidence rates as a function of optical path length difference are plotted for seven different temperatures near the degenerate temperature T_c . For these graphs, data points were taken every 0.20 mm.

From Fig. 5, we find that the dip is deepest at the biphoton source temperature 35.90°C . By finding the temperature at which the visibility will be the greatest, we can then use that temperature to plot our best dip. These data show the uniqueness of this apparatus; measurements for different degeneracies would not be possible for most implementations of HOM interference in an undergraduate laboratory.^{2,3}

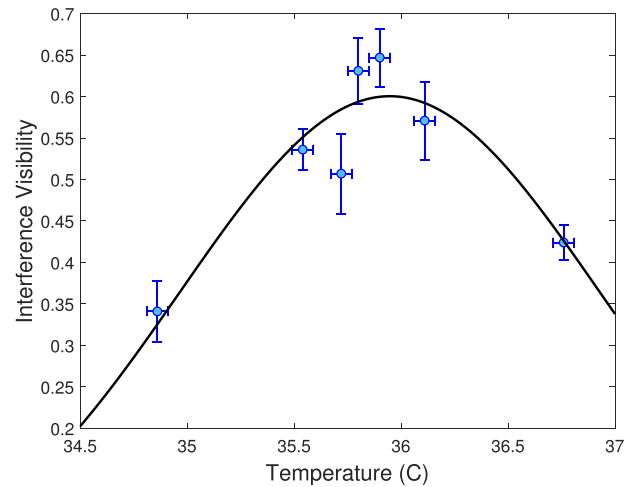


Fig. 6. Dip visibility plotted as a function of PPKTP crystal temperature. These data are assumed to follow Gaussian distribution given in Eq. (8) and are fitted to a model of that form. Horizontal error bars represent estimated crystal temperature fluctuations, vertical error bars represent a 67% confidence interval in dip visibility. The fit has a reduced χ^2 of 1.5 and adjusted R^2 of 0.86.

For a more precise investigation of the relationship between degeneracy and visibility, visibility of the dips is plotted as a function of PPKTP crystal temperature in Fig. 6. This plot allows us to precisely find T_c .

These data in Fig. 6 is fitted to a Gaussian model with a peak at the central temperature T_c . The data in Fig. 6 exemplifies the utility and uniqueness of this apparatus as temperature control is not often found in undergraduate HOM labs. From the fitting data, the central temperature was calculated to be $(35.95 \pm 0.13)^\circ\text{C}$.

V. CONCLUSION

We have presented a versatile and user-friendly apparatus for investigating HOM interference in an undergraduate laboratory. The optical-fiber-based apparatus allows for easier investigation of coherence length and interference visibility for both distinguishable and indistinguishable photon pairs. As an example of the possible experiments that can be done, we showed a relationship between interference visibility and PPKTP crystal temperature.

We demonstrate quantum interference and show that the HOM dip follows an inverted Gaussian profile (Fig. 4). We also find the degenerate temperature T_c and show the effect of crystal temperature on interference visibility.

This apparatus is well suited for implementation in an undergraduate instructional laboratory. The use of fiber optic components makes alignment and setup more intuitive and less tedious for a student who may be less comfortable with experimental optics, while the alignment of the in-air delay lines still gives experience using more traditional optomechanical equipment, making this lab a perfect middle ground between off-the-shelf setups and setups built from the ground up. Furthermore, the use of fiber optics, specifically the fused-fiber couplers, exposes a student to more specialized physical phenomena not commonly included in undergraduate curricula such as the physics of fiber optics and SPDC.²¹

These types of quantum optics experiments for undergraduates are being implemented by other institutions and provide valuable intuition for students regarding the behavior of quantum mechanics.²² Although quantum interference phenomena are common examples of the “weirdness” of quantum mechanics,²³ undergraduates rarely get to see the specific physical characteristics of these phenomena. This apparatus provides that opportunity through the study of HOM interference.

In this paper, we investigated the relationship between temperature and visibility; however, there are other possible experiments that can be done with our setup. We have a clear method of measuring the coherence length of the interference by measuring the FWHM. The relationship and possible dependency of coherence length on the various parameters of interference can be investigated.

Another possible experiment involves the polarization of the interfering photons. Here, we have exclusively scanned over optical path length difference to create the HOM dips. However, it is equally valid to consider differences in polarization between the photon pairs instead of the path length.¹⁶ The behavior of the HOM dips with respect to polarization would be interesting to investigate using this apparatus and would require careful consideration of how the changes in polarization affect the behavior of the polarization-maintaining fused-fiber coupler.

SUPPLEMENTARY MATERIAL

Please click on [this link](https://doi.org/10.60893/figshare.ajp.c.7510881) to access the supplementary material for a list of parts and their respective costs. Print readers can see the supplementary material at <https://doi.org/10.60893/figshare.ajp.c.7510881>.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflict of interest to disclose.

APPENDIX: ACTION OF THE BEAMSPLITTER MATRIX ON THE TWO-PHOTON STATE

While this math is not vitally important to understanding this paper, it is helpful for students less familiar with these types of probability distributions to get a clearer view of what is going on. When our photons pass through the fused-fiber coupler, we are in fact acting on the state of Eq. (2) with the matrix operator $\hat{M} \otimes \hat{M}$, where $\hat{M}$ is the matrix operator of Eq. (1). Although our two-photon state is a more complicated integral over different wavelengths, we can demonstrate the action of $\hat{M} \otimes \hat{M}$ using a much simpler state.

Suppose instead of the state of Eq. (2), our two-photon system only has one wavelength. Then it would be represented by

$$|\Psi_{in}\rangle = |1\rangle|2\rangle = |1\rangle \otimes |2\rangle, \quad (\text{A1})$$

where $|1\rangle$ is the photon entering input 1, and $|2\rangle$ is the other photon entering input 2. The action of the coupler operator is much simpler and gives us

$$\begin{aligned} (\hat{M} \otimes \hat{M})|1\rangle|2\rangle &= (\hat{M}|1\rangle) \otimes (\hat{M}|2\rangle) \\ &= \begin{pmatrix} t & r \\ -r & t \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} t & r \\ -r & t \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} t \\ -r \end{pmatrix} \otimes \begin{pmatrix} r \\ t \end{pmatrix} \\ &= (t|1\rangle - r|2\rangle) \otimes (r|1\rangle + t|2\rangle) \\ &= tr|1\rangle|1\rangle + t^2|1\rangle|2\rangle - r^2|2\rangle|1\rangle - tr|2\rangle|2\rangle. \end{aligned} \quad (\text{A2})$$

When we assume that the two photons are indistinguishable, we get

$$|1\rangle|2\rangle = |2\rangle|1\rangle, \quad (\text{A3})$$

and Eq. (A2) becomes

$$(\hat{M} \otimes \hat{M})|1\rangle|2\rangle = tr|1\rangle|1\rangle + (t^2 - r^2)|1\rangle|2\rangle - tr|2\rangle|2\rangle. \quad (\text{A4})$$

Note that when $t = r = \sqrt{2}/2$, $t^2 - r^2 = 0$, the only possible outcomes are $|1\rangle|1\rangle$ and $|2\rangle|2\rangle$, giving perfect HOM interference and no coincidences, since coincidence only occurs when the photons exit through different outputs.

However, when $r \neq t$, calculating probabilities for each outcome is still quite simple. Since r and t are real numbers, we merely square the coefficient of each state in the wavefunction to find its probability.

Therefore, the probability of a coincidence would be $(t^2 - r^2)^2$ and the probability of not measuring a coincidence would be $2t^2r^2$. These calculations are in fact exactly what is happening behind the scenes when we evaluate Eq. (4) to find the probabilities of different outcomes. The only difference is, this type of calculation is done at every possible wavelength pairing for the two photons and then integrated over all wavelengths. A more detailed treatment of this mechanism as well as other mathematical properties of a beamsplitter can be found in Makarov, 2022.¹¹

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¹Chong-Ki Hong, Zhe-Yu Ou, and Leonard Mandel, "Measurement of sub-picosecond time intervals between two photons by interference," *Phys. Rev. Lett.* **59**(18), 2044–2046 (1987).

²Nicholas S. DiBrito and Enrique J. Galvez, "An easier-to-align Hong–Ou–Mandel interference demonstration," *Am. J. Phys.* **91**(4), 307–315 (2023).

³A. Ourjountsev, M. C. Dheur, T. Avignon, and L. Jacubowicz, "Two-photon quantum interference for an undergraduate lab," *Eur. J. Phys.* **36**(6), 065034 (2015).

⁴Jorge Carvioto-Lagos, Gustavo Armendariz, Víctor Velázquez, Enrique López-Moreno, M. Grether, and E. J. Galvez, "The Hong–Ou–Mandel interferometer in the undergraduate laboratory," *Eur. J. Phys.* **33**(6), 1843–1850 (2012).

⁵Alessandro Fedrizzi, Thomas Herbst, Andreas Poppe, Thomas Jennewein, and Anton Zeilinger, "A wavelength-tunable fiber-coupled source of narrowband entangled photons," *Opt. Express* **15**(23), 15377–15386 (2007).

⁶K. S. Thyagarajan and Ajoy Ghatak, *Fiber Optic Essentials* (John Wiley & Sons, 2007).

⁷Thiadmer Riemersma, "Termite: A simple RS232 terminal" (2019), see https://www.compuphase.com/software_termite.htm.

⁸Gregor Weihs, Michael Reck, Harald Weinfurter, and Anton Zeilinger, "Two-photon interference in optical fiber multiports," *Phys. Rev. A* **54**(1), 893–897 (1996).

⁹Bishnu Pal, "Fabrication and modeling of fused biconical tapered fiber couplers," *Fiber Integr. Opt.* **22**(2), 97–117 (2003).

¹⁰Enrique Peacock-López, "Exact solutions of the quantum double square well potential," *Chem. Educ.* **11**, 383–393 (2006).

¹¹Dmitry Makarov, "Theory for the beam splitter in quantum optics: Quantum entanglement of photons and their statistics, HOM effect," *Mathematics* **10**(24), 4794 (2022).

¹²Marco Fiorentino, Sean M. Spillane, Raymond G. Beausoleil, Tony D. Roberts, Philip Battle, and Mark W. Munro, "Spontaneous parametric down-conversion in periodically poled KTP waveguides and bulk crystals," *Opt. Express* **15**(12), 7479–7488 (2007).

¹³G. Dattoli, A. Renieri, and A. Torre, "Generalized polynomials and derivation of integrals involving products of Hermite polynomials," *Radiat. Phys. Chem.* **53**(4), 391–395 (1998).

¹⁴Baghdasar Baghdasaryan, Fabian Steinlechner, and Stephan Fritzsche, "Maximizing the validity of the Gaussian approximation for the biphoton state from parametric down-conversion," *Phys. Rev. A* **106**(6), 063714 (2022).

¹⁵M. V. Fedorov, Yu. M. Mikhailova, and P. A. Volkov, "Gaussian modeling and Schmidt modes of SPDC biphoton states," *J. Phys. B* **42**(17), 175503 (2009).

¹⁶Richard A. Campos, Bahaa E. A. Saleh, and Malvin C. Teich, "Fourth-order interference of joint single-photon wave packets in lossless optical systems," *Phys. Rev. A* **42**(7), 4127–4137 (1990).

¹⁷Qubitekk, see <https://www.compadre.org/advlabs/bfyiii/files/QubitekkBrochureDigital05122017final.pdf> for "Quantum Mechanics Education and Research" (2017).

¹⁸Fabian Steinlechner, Marta Gilaberte, Marc Jofre, Thomas Scheidl, Juan P. Torres, Valerio Pruneri, and Rupert Ursin, "Efficient heralding of polarization-entangled photons from type-0 and type-ii spontaneous parametric downconversion in periodically poled KTiOPO₄," *J. Opt. Soc. Am. B* **31**(9), 2068–2076 (2014).

¹⁹Brett J. Pearson and David P. Jackson, "A hands-on introduction to single photons and quantum mechanics for undergraduates," *Am. J. Phys.* **78**(5), 471–484 (2010).

²⁰Dong-Yeop Na and Weng Cho Chew, "Classical and quantum electromagnetic interferences: What is the difference?," *Prog. Electromagn. Res.* **168**, 1–13 (2020).

²¹National Research Council, *Adapting to a Changing World: Challenges and Opportunities in Undergraduate Physics Education* (National Research Council, 2013).

²²Enrique J. Galvez, "Resource letter SPE-1: Single-photon experiments in the undergraduate laboratory," *Am. J. Phys.* **82**(11), 1018–1028 (2014).

²³William J. Mullin, *Quantum Weirdness* (Oxford U. P., 2017).